MILITARY SPECIFICATION SHEET

ELECTRON TUBE, POWER
TYPE 8116

This specification is approved for use by Naval Electronic Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the electron tube described herein shall consist of this document and the latest issue of Specification MIL-E-1.

DESCRIPTION: Double tetrode
See figure 1
Mounting position - - - Vertical, base up or down; horizontal with anode pins in a horizontal plane.
Weight - - - - - - - - 2.5 ounces nominal (70.9 grams)

ABSOLUTE RATINGS:

Parameter:	Ef	F	Eb	Ec1	Ec2	Ehk	Ib	Ic1
Unit:	Vac	MHz	Vdc	Vdc	Vdc	Vdc	mAdc	mAdc
Maximum:								
AB, SSE (Note 2)	13.25/26.5 ±10%	30	1,000	-175	360	±100	220	10
TEST CONDITIONS:	13.25/26.5	---	600	Adj	250	0	40	---

ABSOLUTE RATINGS:

Parameter:	Pg1	Pg2	Pp	Pk	T	tk
Unit:	W	W	W	W	°C (Note 3)	sec (Min)
Maximum:						
AB, SSB (Note 2)	2x1	7	2x30	200	---	30
TEST CONDITIONS:	---	---	---	---	---	120

GENERAL:

Qualification - Not required

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METHOD	REQUIREMENT OR TEST	NOTES	CONDITIONS	AQL (PERCENT DEFECTIVE)	INSPECTION LEVEL OR CODE	SYMBOL	LIMITS MIN	LIMITS MAX	UNIT
	Quality conformance inspection, part 1								
1231	Peak emission	5, 6	eb = ec1 = ec2 = 225 v	1.0	II	is	1.5	---	a
1256	Electrode current (screen)	5		1.0	II	Ic2	---	6.0	mAdc
1261	Electrode voltage (1) (grid)	5		1.0	II	EC1	-25	-35	Vdc
1266	Total grid current	1, 5	Ib = 60 mAdc; t = 180	1.0	II	Ic1	---	-6.0	µAdc
1301	Heater current	-	If = 13.25 V; t = 300	1.0	II	If	760	1,260	mAac
1201	Short and discontinuity detection	-		0.65	II	---	---	---	---
	Quality conformance inspection, part 2								
1031	Low-frequency vibration	5	Ec1/Ib = 10 mAdc; Eb = 250 Vdc; Rp = 2,000 ohms	---	---	---	---	2,000	mVac
1036	Bump	-	Angle = 30°	---	---	---	---	---	---
1261	Electrode voltage (2) (grid)	9		---	---	Ec1	-40	-52	Vdc
1256	Electrode current (anode)	5	Ec1 = -85 Vdc	---	---	Ib	---	10	mAdc
---	Power output (1)	3, 4 8, 10	Paralleled operation class AB1 amplifier; Ef = 26.5 Vac; F = 75 Mhz (max); Eb = 1,000 Vdc; Ec1/Ib = 55 mAdc; Ec2 = 270 Vdc; Ic2 = 30 mAdc (max)	---	---	Po	130	---	w (useful)
---	Power output (2)	-	After 5 minutes, power output (1)	---	---	ΔPo/Δt	---	-0	%
---	Power output (3)	-	Ef = 23.9 Vac; power output (1)	---	---	ΔPo/ΔEf	---	15	%
1331	Direct-interelectrode capacitance	7		---	---	Cgp Cin Cout	0.050 9.4 2.6	0.080 11.8 3.7	pF pF pF
	Quality conformance inspection, part 3								
---	Life test	3, 8	Group C; power output (1); t = 500 hours except; Ec1/Ib = 50 mAdc; Ec2 = 250 Vdc	---	---	---	---	---	---
---	Life-test end points:								
1231	Peak emission	-		---	---	is	1.1	---	a
---	Power output (1)	-		---	---	Po	120	---	w (useful)
1256	Electrode current (grid)	-		---	---	Ic1	---	-100	µAdc
1031	Vibration	11	F = 50 Hz; t = 900; no voltages	---	---	---	---	---	---

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NOTES:

1. This test is to be the first test performed at the conclusion of the holding period.
2. Both sections in parallel.
3. Maximum glass and seal temperature shall be 250°C. When operating the tube at maximum ratings, with forced air cooling, it is necessary to cool both the lateral bulb surface as well as the anode pin seals by directing the flow of air toward the top and sides of the bulb. In most cases, approximately 20 cfm is sufficient. At reduced ratings, the use of a heat sink is permissible. However, regardless of the type of cooling being used, the degree of cooling shall be determined by direct temperature measurement of both the seals and the bulb.
4. Power output is total power at drive frequency delivered to load. Drive power is not subtracted from total output. Output circuit tuned for maximum power output.
5. Read each unit separately. Grid (1) of section not under test shall be connected to -100 Vdc.
6. Tie both anodes and the screen together. The pulse duration measured at 5 percent of the maximum value shall be not less than 3 µs. At 50 percent amplitude, the duration shall be less than 2 µs. The applied voltage shall have a maximum repetition rate such that the duty cycle, based on the pulse length measured at 50 percent amplitude, shall not exceed 0.0002 (0.02 percent).
7. The test shall be duplicated on each unit separately. The unit not under test shall be tied to ground. Shielding shall be in accordance with figure 2.
8. Operation as class "AB1" amplifier, Q = 10 minimum. Power output is total power at drive frequency delivered to load. Drive power is not subtracted from total output. Output circuit tuned for maximum power output. Drive to true "0" bias point condition which shall be met after all tuning is completed.
9. Ef = 13.25, Eb = 800 Vdc, Ec2 = 325 Vdc, anodes and grids (1) in parallel, vary Ec1 for Ib = 45 mAdc (total, both sections).
10. These test conditions exceed the absolute maximum ratings and operation shall limited to 5 minutes maximum "on" time followed by an "off" time of at least 5 minutes. Total "on" time for power output (1) plus power output (2) tests shall be limited to 5 minutes with power output (3) test requiring as short a time as possible beyond the 5 minutes to reach stabilization.
11. Vibration test shall be checked annually, using a sample of three tubes with no failures allowed. In case of a sample failure, the test shall become a part of quality conformance inspection (QCI), part 2, for three consecutive successful submissions, at which time the test may revert to the QCI, part 3 test.
12. Revision letter symbols are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

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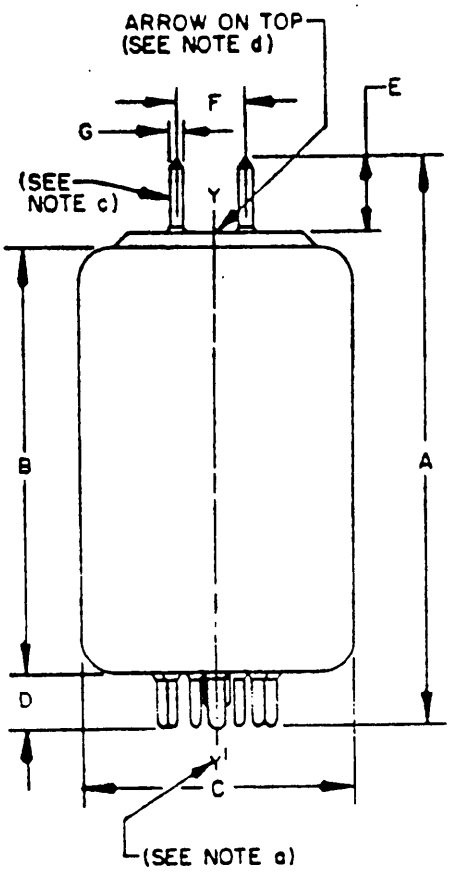
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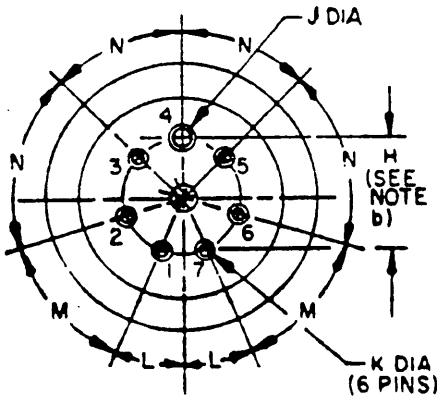
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Ltr	Dimensions in inches with metric equivalents (mm) in parentheses		Notes
	Minimum	Maximum	
Quality conformance inspection, part 2			e
A	3.594 (91.29)	4.031 (102.39)	
B	2.688 (68.28)	2.938 (74.63)	
C	1.710 (43.43)	1.740 (44.20)	
D	.375 (9.53)	.500 (12.70)	
E	.531 (13.49)	.594 (15.09)	
F	.722 (18.34)	.738 (18.75)	c
G	.122 (3.10)	.128 (3.25)	
K	.052 (1.32)	.060 (1.52)	
Reference dimensions			
H	1.000 (25.40)		b
J	.125 (3.18)		
L	26°		
M	52°		
N	51°		



Pin connections	
Pin No.	Element
1	h
2	2g1
3	g2
4	k
5	hct
6	1g1
7	h

FIGURE 1. Outline drawing of electron tube type 8116.

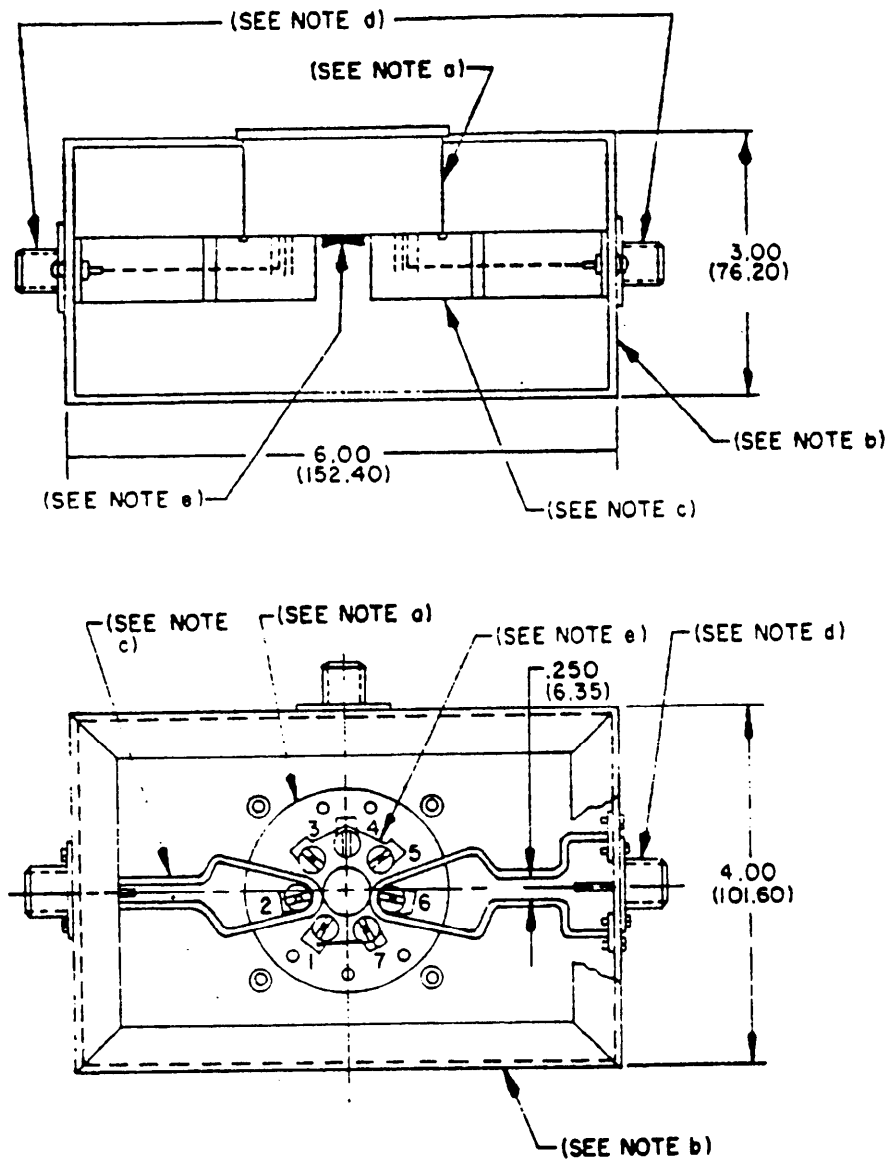
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NOTES:

a. The axis Y-Y1 is defined as the axis of the base pin gage described in note b.
b. The tube base shall be capable of entering to a distance of .375 (9.53 mm) a flat plate gage having six holes .0800 (2.03 mm) ±.0005 (.01 mm) and one hole .1450 (3.68 mm) ±.0005 (.01 mm), all arranged on a 1.0000 (25.40 mm) ±.0005 (.01 mm) circle at the angles ±5' specified on the outline. A .500 (12.70 mm) ±.010 (.25 mm) diameter hole at the center of the pin circle shall also be required. The axis Y-Y1 is defined by the center of this hole.
c. The anode leads shall be capable of entering a flat plate gage of .375 (9.53 mm) ±.001 (.03 mm) thickness having two holes .2000 (5.08 mm) ±.0005 (.01 mm) diameter spaced .365 (9.27 mm) ±.001 (.03 mm) from a point coincident with the axis Y-Y1. The axes of the holes shall be parallel to axis Y-Y1 and the plane of these axes shall be 90 degrees ±5' from the plane through axis Y-Y1 and pin No. 4.
d. Location of large base pin shall be referenced by an arrow in the top stem.
e. Tube shall be inserted in jig shown on figure 3. Tubes that do not fit in the jig shall be rejected.

FIGURE 1. Outline drawing of electron tube type 8116 - Continued.

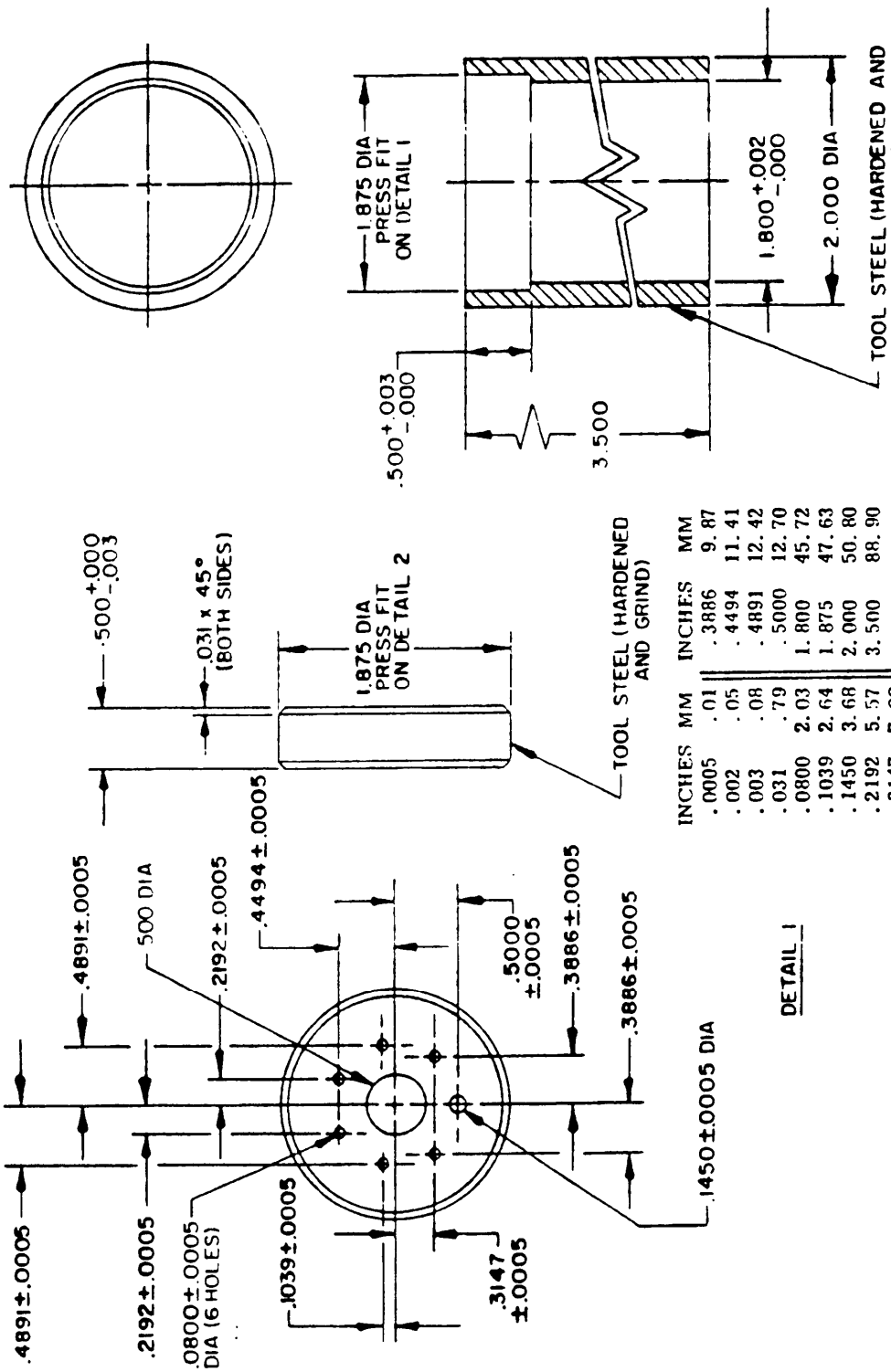
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NOTES:
a. Septar tube socket (7 pin) (Johnson No. 122-101-100 or equivalent).
b. Aluminum mounting box 6 (152.40 mm) x 4 (101.60 mm) x 3 (76.20 mm).
c. Copper shielding, .040 (1.02 mm) thick x .750 (19.05 mm) wide. (May be hand formed to shape.)
d. Female sockets for RF connectors (Amphenol SO-239 or equivalent).
e. Copper connection wire, AWG No. 20.
f. Metric equivalents are in parentheses.

FIGURE 2. Mounting box with 7 pin Septar socket, shielding, and connection lead.

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INCHES	MM	INCHES	MM
.0005	.01	.3886	9.87
.002	.05	.4494	11.41
.003	.08	.4891	12.42
.031	.79	.5000	12.70
.0800	2.03	1.800	45.72
.1039	2.64	1.875	47.63
.1450	3.68	2.000	50.80
.2192	5.57	3.500	88.90
.3147	7.99		

NOTES:
a. Alternate method: May use standard drill bushing press fit in position.
b. Pin circle to be concentric to 1.875 (47.63 mm) dia within .002 (.05 mm) TIR.
c. Dimension 1.875 (47.63 mm) to be concentric to 1.800 (45.72 mm) dia within .002 (.05 mm) TIR.

FIGURE 3. Outline gage for electron tube type 8116.

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